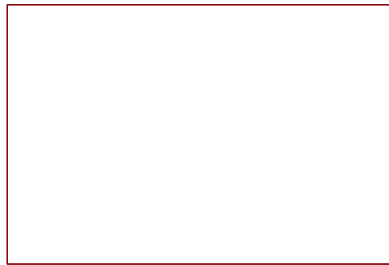


Root [0]

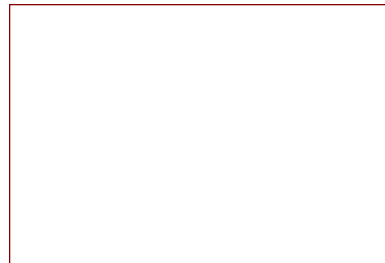
Technical Sheet Only!
Please refer to the subsequent sheets for schematic

Power



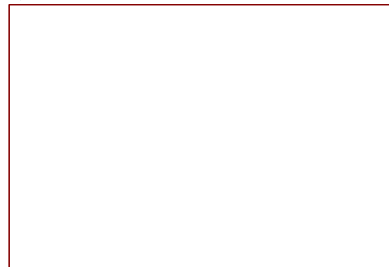
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Sensor



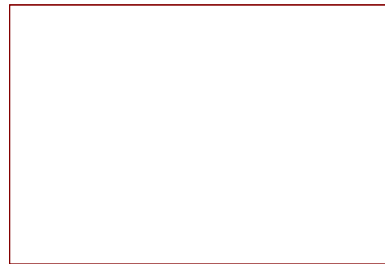
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System



File: System.kicad_sch

eFuse



File: efuse.kicad_sch

Connectivity



File: connectivity.kicad_sch

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Sheet: /

File: VoxSense.kicad_sch

Title: VoxSense Sensor Board – Root

Size: A4

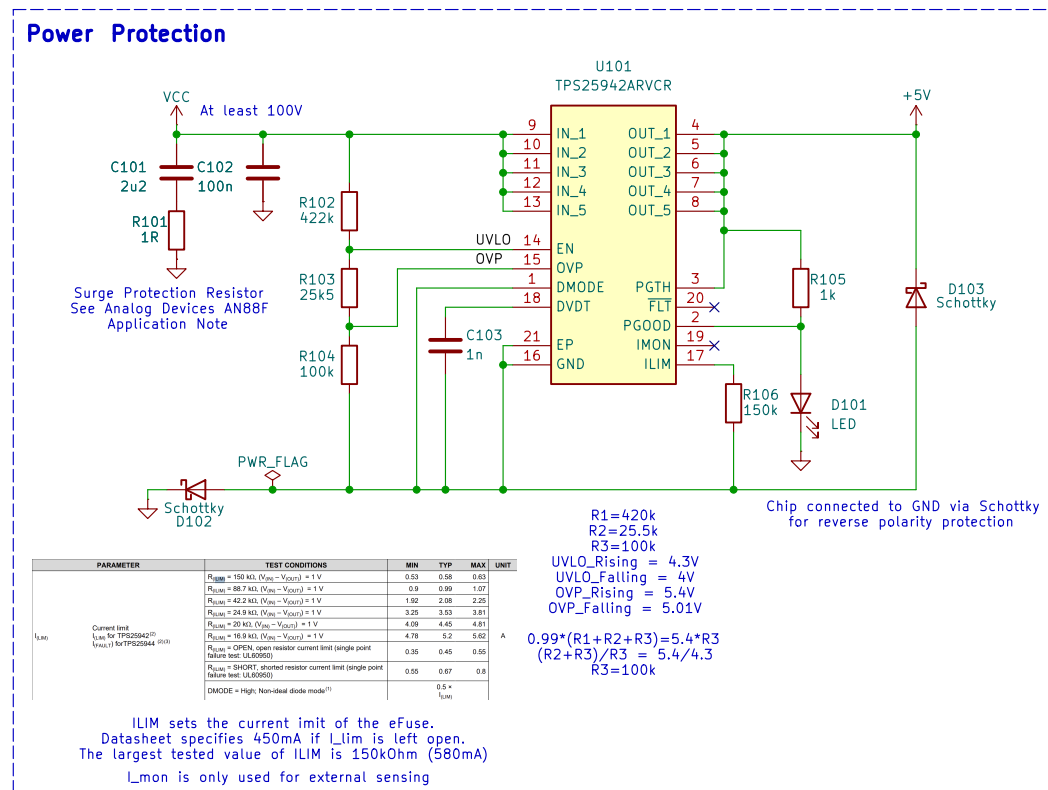
Date: 2025–11–17

Rev: 1.0

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Id: 0/6

eFuse [1]



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Sheet: /eFuse/
File: efuse.kicad_sch

Title: eFuse

Size: A4	Date: 2025-12-04
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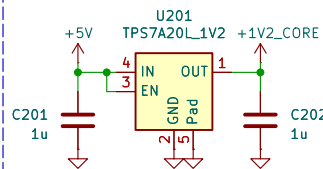
Size: A4	
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Rev: 1.0

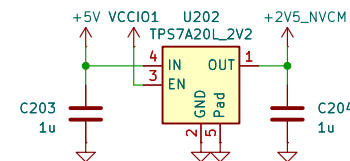
Id: 1/6

Power [2]

Core Supply Voltage



NVCM Supply



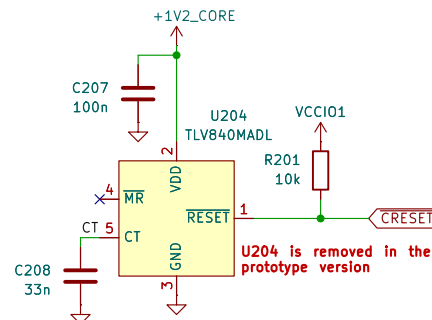
The NVCM regulator can be omitted and the VPP2V5 pin can be connected directly to 3V3 in order to save cost.

The pin can handle 3.6V.

A diode can be used to drop the voltage before by roughly 0.7V

FPGA Power Delay

Note that CRESET is active LOW, meaning the FPGA is active when CRESET is HIGH. This requires a RESET variant of the TLV840 chip which active LOW, meaning it is the correct reset logic.

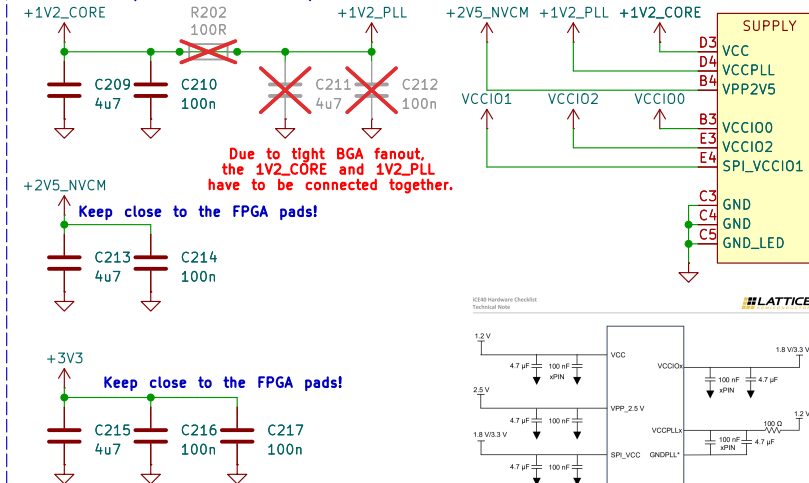


$$C [\mu F] = (T_{\text{delay}} [\mu S] - 80\mu S) / 618937$$
$$32nF = (20,000\mu s - 80\mu S) / 618937$$

The FPGA wakes up after roughly 10us, however, the memory takes 150us to boot. We need to delay the FPGA startup, so that the flash memory is ready.

FPGA Power Input

Keep close to the FPGA pads!



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Technical Notes

LATTICE

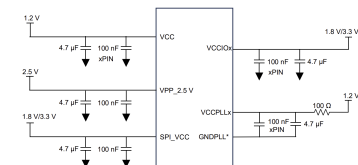
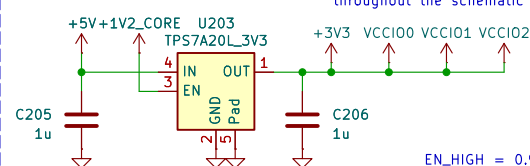


Figure 2.1: Typical Power Supply Filter
Filtering suggested by the manufacturer

IO Supply



Multiple net names are used to indicate the I/O banks throughout the schematic

EN_HIGH = 0.9V
EN_LOW = 0.3V
EN_MAX = 6.5V

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Sheet: /Power/

File: power.kicad_sch

Title: Power

Size: A4

Date: 2025-11-09

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Rev: 1.0

Id: 2/6

PicoBlade Programming Connector

FLASH_CS_NOESD
FLASH_MOSI_NOESD
FLASH_MISO_NOESD
FLASH_SCK_NOESD
CDONE
CRESET

VCCD
PicoBlade J302
D303 TVS
D304 TVS
WARNING Connector is FLIPPED to match polarity

Open-Drain Outputs
Used for LED driving

VCC100 = 3V3

VCC100 VCCIO0 VCCIO0

R306 510R R307 510R R308 510R

D305 LED D306 RX D307 TX

BANK 0
OPEN DRAIN

RGB0 A2 LED
RGB1 B1 RX_LED
RGB2 A1 TX_LED
BARCODE A5
IRLED A6

VCCD VCCD

C301 100n

D301 TVS

J301 TC2030-NL

VCCD VCCD

D302 Schottky

D302 is removed in the prototype version

TPD4E02B04 U302

R301 4x22R

ATmega328P

FLASH_CS_NOESD

FLASH_MOSI_NOESD

FLASH_MISO_NOESD

FLASH_SCK_NOESD

FLASH_CS

FLASH_MOSI

FLASH_MISO

FLASH_SCK

ESD Protection IC

External Oscillator

The diagram shows the external oscillator circuit for the Y301 12MHz oscillator. The circuit includes a 100nF capacitor (C303) connected to VCC102 and ground. The Y301 oscillator is connected to VCC102 (pin 1), ground (pin 2), and its output (pin 3) is connected to a 22R resistor (R302) and then to the 12MHZ_CLK input of the Y301.

Part choice is subject to change in the next revision

VCCIO1 = 3V3

Chip is active when $\overline{\text{CRESET}}$ is LOW

CDONE is an Open-Drain Output which goes high after the device has been configured successfully

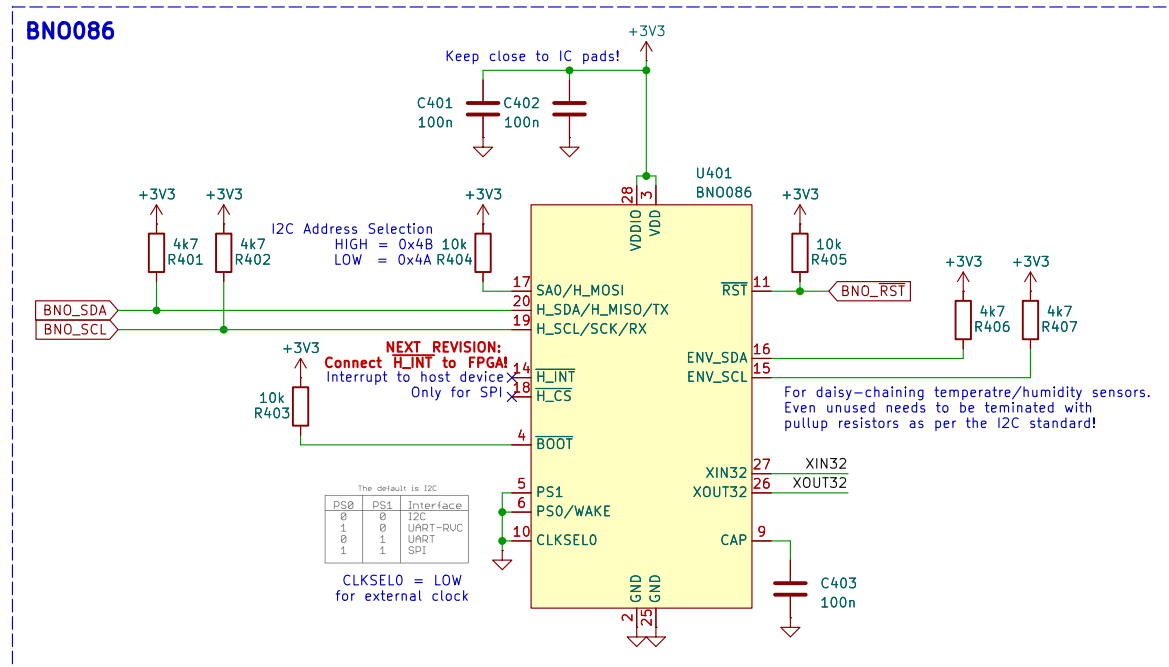
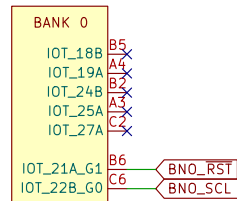
See: ICE40 Programming and Configuration app note
Section 9.2. SPI PROM Requirements

If the LED shines, the device has been configured properly

Id: 3/6

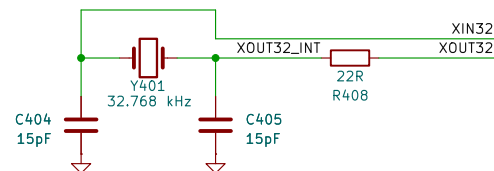
Sensor [4]

I2C Interface



BN0086 Crystal

CEVA recommends using a crystal with tolerance 50ppm with 12.5pF capacitor loading.



$(12.5\text{pF Load Capacitance} - 5\text{pF Stray Capacitance}) * 2 = 15\text{pF Load Capacitors}$

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Sheet: /Sensor/

File: Sensor.kicad_sch

Title: Sensor

Size: A4

Date: 2025-11-18

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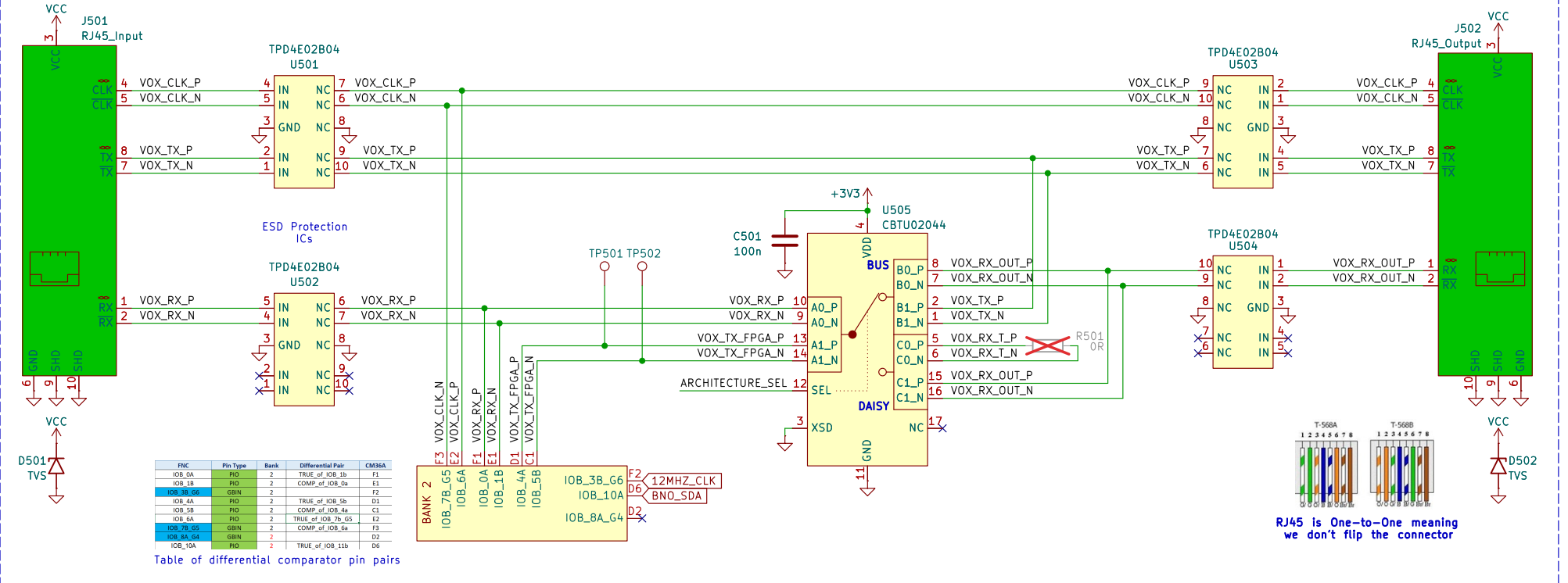
Rev: 1.0

Id: 4/6

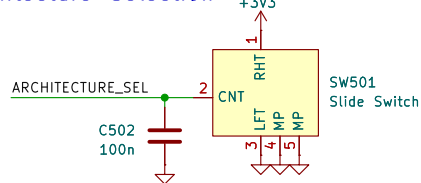
Connectivity [5]

VoxLink Ethernet Connection + Bus / Daisy Chain Switch

No Placement (NOP) – Used for the possibility of terminating the differential pairs to prevent reflections

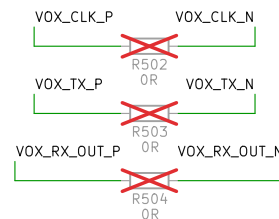


Architecture Selection



SEL = LOW -> A and B are connected
SEL = HIGH -> A and C are connected

Termination Resistors



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Sheet: /Connectivity/
File: connectivity.kicad_sch

Title: Connectivity

Size: A4 Date: 2025-12-21
KiCad E.D.A. 10.0.0

Rev: 1.0
Id: 5/6